

Thesis Topic

Federated Causal Discovery with Probabilistic Circuits: Towards Scalable, Privacy-Preserving Structural Learning

Author

Mei-Ling Fang (TU ID: ge86jin, registration no. 03763921, julie.fang@tum.de),
Devendra Singh Dhami (d.s.dhami@tue.nl)

Abstract

This thesis investigates the integration of probabilistic circuits, such as Sum-Product Networks (SPNs), into federated causal discovery frameworks. Building on the FedCDH method for federated causal discovery proposed by Li et al. (2024), this work incorporates Fast Probabilistic Circuits (FPC) techniques as developed by Seng et al., alongside RAT-SPN architectures introduced by Peharz et al. The research replaces traditional summary statistic based conditional independence testing with a graphical and computational model-based approach to better capture complex dependencies in federated environments. It extends the typical horizontal data split setting to include vertical and hybrid splits specifically designed to take advantage of the einsum network architecture. Experimental results demonstrate that combining FedCDH with FPC and RAT-SPN advances interpretable, privacy-respecting causal structure learning applicable to biomedical and semiconductor domains, supporting collaborative discovery while preserving data sovereignty.

References

- Li, L., Ng, I., Luo, G., Huang, B., Chen, G., Liu, T., Gu, B., & Zhang, K. (2024). Federated Causal Discovery from Heterogeneous Data. *ICLR 2024*. arXiv:2402.13241.
- Seng, K., Liu, T., & Domingos, P. (2023). FPC: Fast Probabilistic Circuits for Scalable Probabilistic Inference. *AAAI Conference on Artificial Intelligence*.
- Peharz, R., Lang, S., Vergari, A., Stelzner, K., Molina, A., Trapp, M., Van den Broeck, G., Kersting, K., & Ghahramani, Z. (2025). Einsum Networks: Fast and Scalable Learning of Tractable Probabilistic Circuits. *ICML 2020, Journal of Machine Learning Research*. arXiv:2004.06231.